# CFPB (Consumer Financial Protection Bureau) Consumer Complaints Dashboard (Power BI)

**Project:** CFPB Consumer Complaints Dashboard (Power BI)
**Goal:** Build a Power BI dashboard analyzing consumer complaints (Product, Trend over time, State, Company).

**File size:** ~702 MB (Kaggle lists it as 863.14 MB uncompressed, but actual downloaded file is ~702 MB)
**Row count (as first reported by Power BI):** 1,970,829 rows, 18 columns – later found to be inaccurate; true count was 1,282,355 (see Resolution below)
**Data range:** Complaints from November 2011 to May 2019
**Source:** CFPB Consumer Complaint Database, mirrored on Kaggle
**Columns include:** Date received, Product, Sub-product, Issue, Sub-issue, Consumer complaint narrative (mostly null — 60-80% null across categories per Kaggle's stats), Company public response, Company, State, ZIP code, Tags, and more (18 total)

**Problem discovered:** When loading `rows.csv` into Power BI:

1. The "Date received" column showed thousands of conversion errors.
2. Investigating showed dates in inconsistent formats and, more importantly, garbage values like "Sincerely," "However," and fragments of sentences appearing inside the Date column.
3. Initial theory: the "Consumer complaint narrative" column contains free-text complaints with line breaks inside the text itself (people hit Enter while typing), and these multi-line fields weren't properly wrapped in quotes — causing the CSV parser to treat a mid-sentence line break as the start of a new row, shifting all subsequent columns out of alignment.
4. Confirmed by opening the raw CSV in a text editor and seeing narrative text sitting alone on lines with no other columns.
5. *This theory was later revised — see Resolution below. The file's quoting was actually fine; the problem was on the parsing side.*

**Decision:** Rather than trying to manually patch this inside Power Query (slow, fragile across ~2 million rows ), switch to Python to clean the file first, since Python's CSV/pandas parser handles this more predictably and we can inspect/fix row alignment before Power BI ever sees the file.

**Actions taken:**

- Loaded `rows.csv` in Python using pandas with the Python engine, which correctly handles quoted multi-line fields
- Dropped the "Consumer complaint narrative" column (not needed for dashboard visuals; also the source of row-miscounting in naive parsers)
- Converted "Date received" to proper datetime type

- Exported clean file as `complaints_clean.csv` for Power BI to load.

**Resolution / Key Finding:**
After running the cleaning script, Python reported **0 bad lines skipped** and **0 rows removed for invalid dates** – meaning no data was actually corrupted or lost.

**The true explanation:** the raw file has 1,970,829 *physical lines* (i.e., newline characters), but only **1,282,355 actual complaint records**. The gap (~688,000) comes from complaint narratives that span multiple lines (people pressing Enter while typing their complaint text), properly wrapped in quotes per CSV standard. Power BI's Text/CSV import doesn't handle quoted multi-line fields correctly, so it miscounted these wrapped narrative lines as separate rows – creating fake extra "rows" with misaligned/blank columns and garbage values in Date received.

Python's pandas (with `engine="python"`) correctly parses quoted multi-line fields per the CSV standard, merging each complaint's full narrative back into one row. This gives the accurate record count. In other words, the original file was never malformed; the initial theory was wrong, and the real issue was that Power BI's default parser couldn't handle valid multi-line CSV content.

**Verification:** 1,282,355 matches the expected total when summing Kaggle's published date-range bucket counts (Nov 2011–Nov 2019), confirming this is the correct row count – not a data-loss artifact.

**Final clean file:** `complaints_clean.csv` — 1,282,355 rows, 17 columns (narrative column dropped), Date received standardized to proper date type, ready for Power BI import.

**Dashboard build (Power BI):**

Loaded `complaints_clean.csv` into a fresh Power BI file – loaded cleanly with no errors and the correct row count. Built four visuals:

1. **Card** — Total Complaints (Count Distinct of Complaint ID) = 1,282,355
2. **Clustered Column Chart** — Complaints by Product (Mortgage highest, followed by Debt collection and Credit reporting)
3. **Line Chart** — Complaints trend by Year (2011–2019), showing a steady rise peaking around 2017–2018
4. **Filled Map** — Complaints by State, shaded by complaint volume, with tooltips showing exact counts per state

Applied a built-in Power BI theme for consistent styling, gave each visual a descriptive title, added a page header, and arranged everything into a clean dashboard layout.

**Key lessons learned:**

- Don't trust a tool's row count at face value – verify against an independent source when a number looks off
- Multi-line quoted text fields in CSVs can silently break naive parsers (like Power BI's default Text/CSV import); Python's pandas with the Python engine handles this correctly
- When debugging data issues, inspect the raw file directly in a text editor rather than only troubleshooting inside the target tool
- The first theory about a problem isn't always right — here, the actual root cause was the opposite of the initial assumption: the file was fine, and the importer was the limitation

**Dashboard:**